

In the tangential direction, resultant force acting on the bob = $W \sin \theta$

Hence, the resultant force should be in direction Q.

10 A At the bottom of the hill, the net force = $N - mg = \text{centripetal force} = ma$

Hence, $N = mg + ma > 0$.

Therefore, the car can never lose contact at the bottom of the hill.

At the top of the hill, the net force = $mg - N = ma$

Hence, $N = mg - ma$

Therefore, the car will lose contact (feel weightless) when $N < 0$

$$\Rightarrow g < a$$

$$\Rightarrow g < \frac{v^2}{R}$$

$$\Rightarrow v > \sqrt{gR}$$